Domain Name System (DNS)

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NUS | Computing
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Addressing

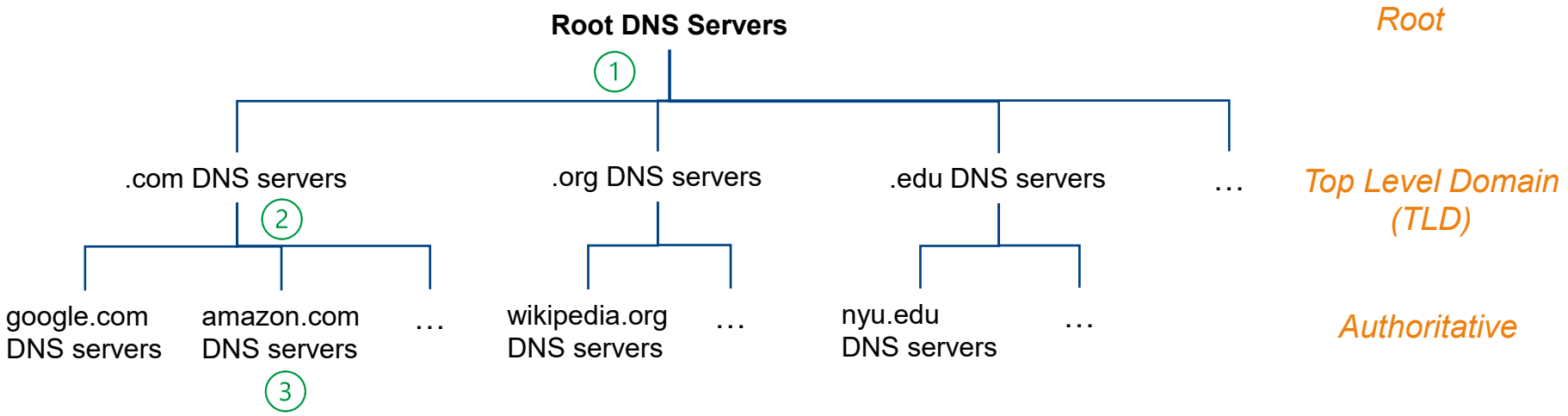
- People: many identifiers – NRIC, name, passport #
- Internet hosts, routers:
  - IP address (32 bit == 4 bytes), e.g., 45.60.35.225 – used for addressing packets
  - "Name", e.g., comp.nus.edu.sg – used by humans
- How to map a name to IP address, and vice versa?

Domain Name System (DNS):

- *Distributed database* implemented as a *hierarchy* of many *name servers*
- An *application-layer protocol* allowing hosts to query the distributed database

Specified in RFC 1034 and 1035, updated in several additional RFCs

DNS: A Distributed, Hierarchical Database



Client wants IP address for www.amazon.com – a simple view:

1. Client queries **root server** to find .com DNS server
2. Client queries **.com DNS server** to get amazon.com DNS server
3. Client queries **amazon.com DNS server** to get IP address for www.amazon.com

Root Name Servers

- 13 named authorities (logical root name) serve the DNS root zone → IANA listing
  - Each with an IP address
  - Each “replicated” many times – has multiple servers to whom requests can be distributed using anycast routing
  - A total of hundreds of servers in many countries
- ICANN (Internet Corporation for Assigned Names and Numbers) manages root DNS domain
- All DNS lookups start at the root zone
  - Every DNS resolver has a list of the root server addresses built in

As of 2026-03-13T10:22:52Z, the root server system consists of 2015 instances operated by the 12 independent root server operators.

TLD and Authoritative DNS Servers

Top-Level Domain Servers

- Responsible for TLD domains (.com, .org, .net, .edu, .gov, …) and all top-level country domains (.sg, .cn, .uk, .ca, .jp, …)
- Administered by authorized domain registrars (accredited by ICANN), e.g.:
  - .com and .net: Verisign (who also operates two root name servers)
  - .edu: Educause

Authoritative DNS Servers

- Organization's own DNS server(s), providing authoritative mappings for its publicly accessible, named hosts
  - E.g., Google: maps.google.com, mail.google.com, …
- Can be maintained by the organization or a service provider

Local DNS Name Servers

To check the local DNS name server:

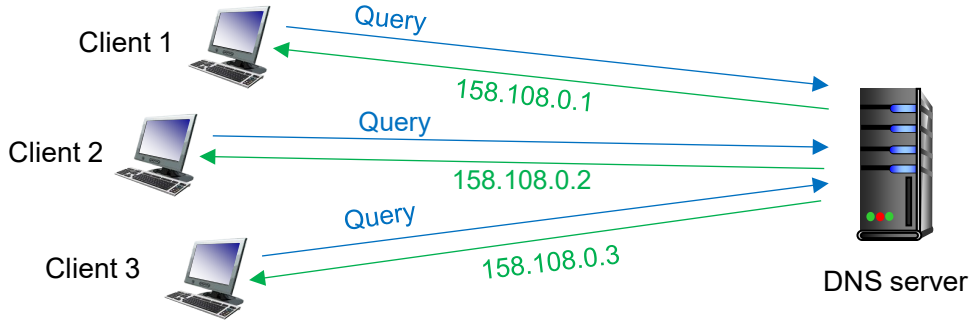
- Windows: `ipconfig /all`
- MacOS: `scutil --dns`

- Does not strictly belong to the hierarchy
- Each ISP provides the hosts (customers) with the address of a local DNS server (a.k.a. default name server)
  - For residential customers, this can be the router

- When a host (browser) wants to request an URL, it first needs to make a DNS query to map the name to the IP address
- The host sends the DNS query to the local DNS server
- The local DNS server acts as a proxy to forward the query into the DNS server hierarchy for resolution

DNS Services

- Hostname-to-IP-address translation
- Host aliasing
  - *Canonical hostname* e.g., `relay1.west-coast.enterprise.com`
  - *Alias names* e.g., `enterprise.com` and `www.enterprise.com`
- E-mail server aliasing
- Load distribution
  - Busy sites can be replicated over multiple servers, each with a different IP address
  - The authoritative name server uses a balancing strategy to select an address when queried, e.g., round-robin

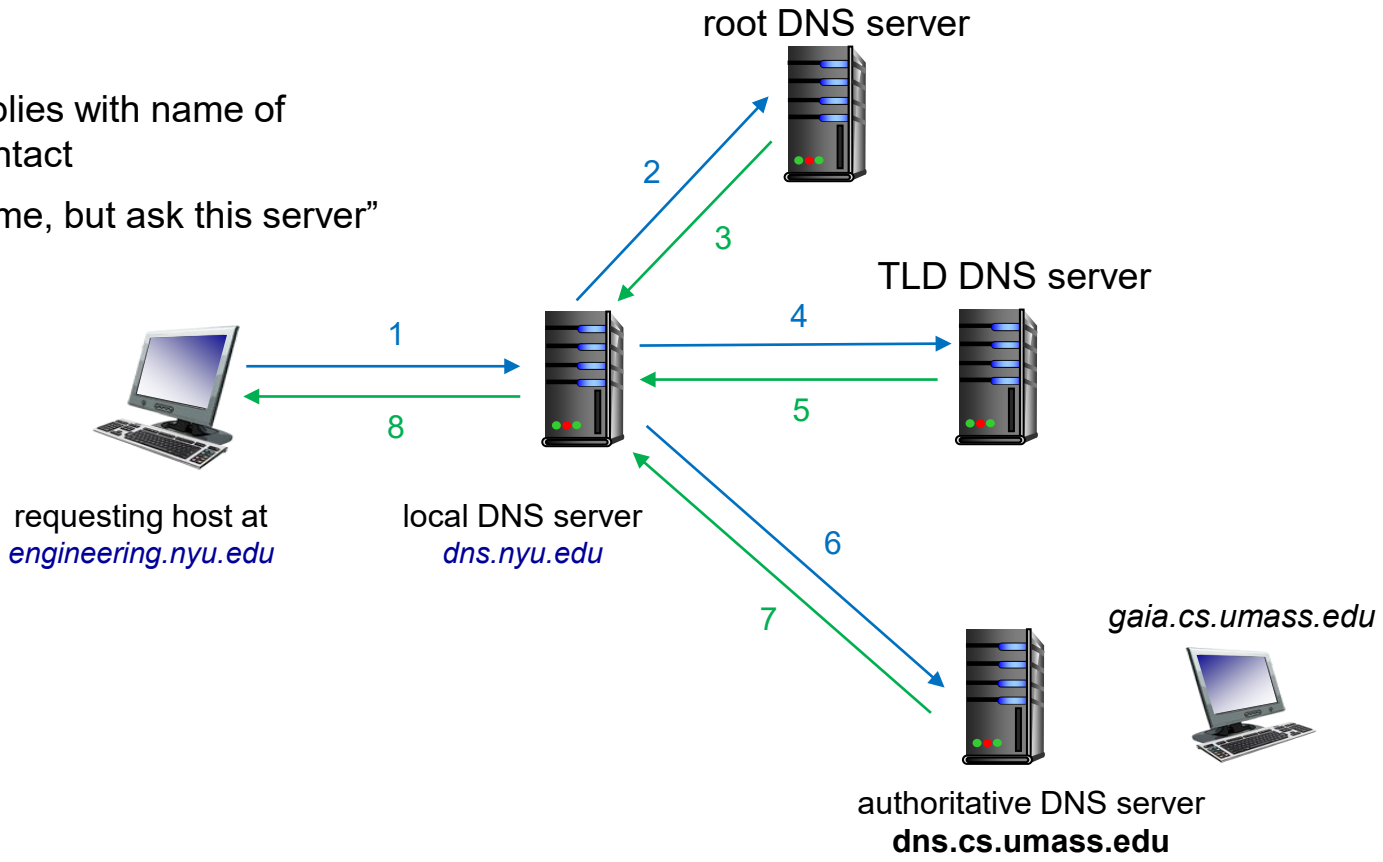


Hostname	IP Address
relay1.west-coast.enterprise.com	158.108.0.1
relay1.west-coast.enterprise.com	158.108.0.2
relay1.west-coast.enterprise.com	158.108.0.3

DNS Name Resolution: Iterated Query

Iterated query:

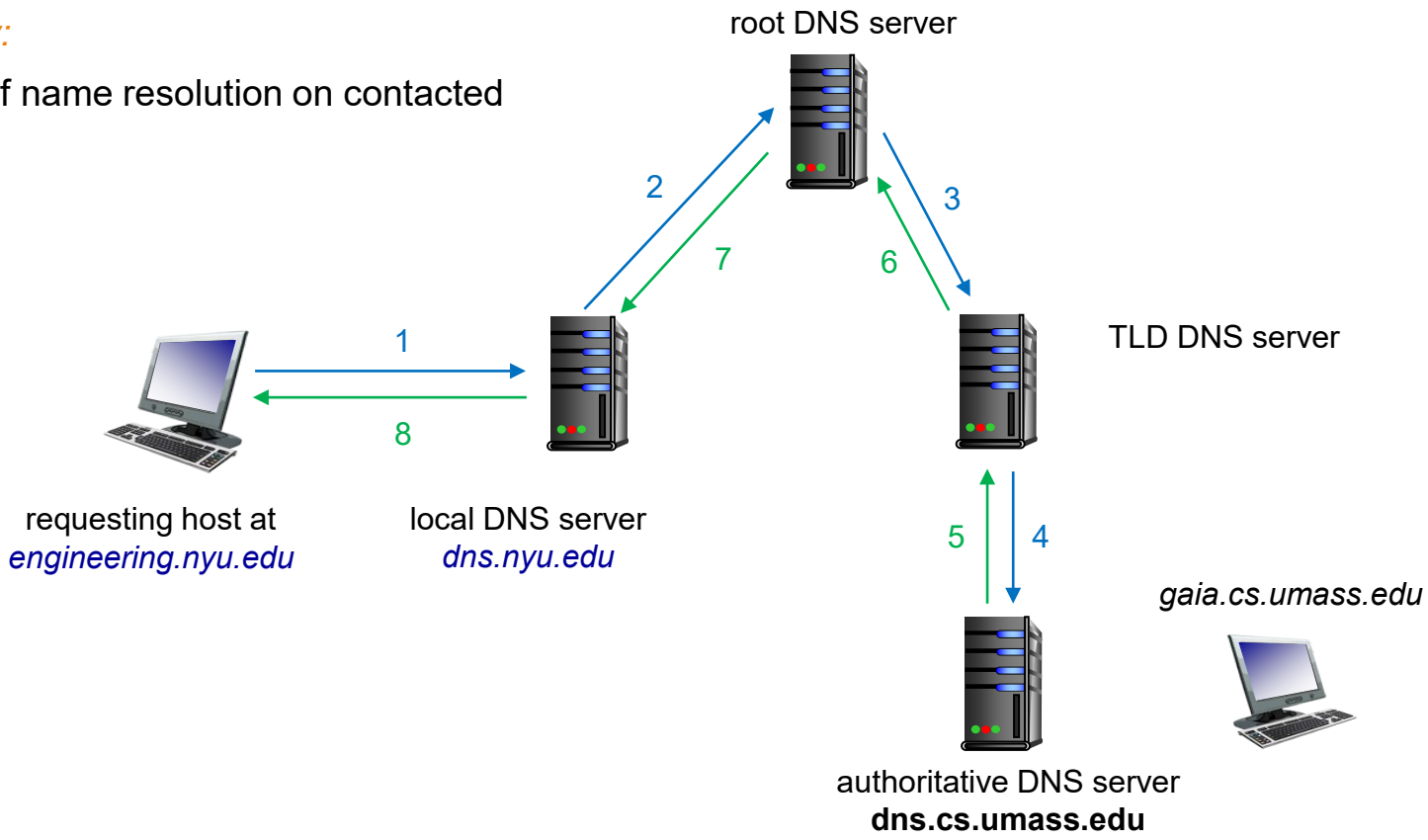
- Contacted server replies with name of another server to contact
- "I don't know this name, but ask this server"



DNS Name Resolution: Recursive Query

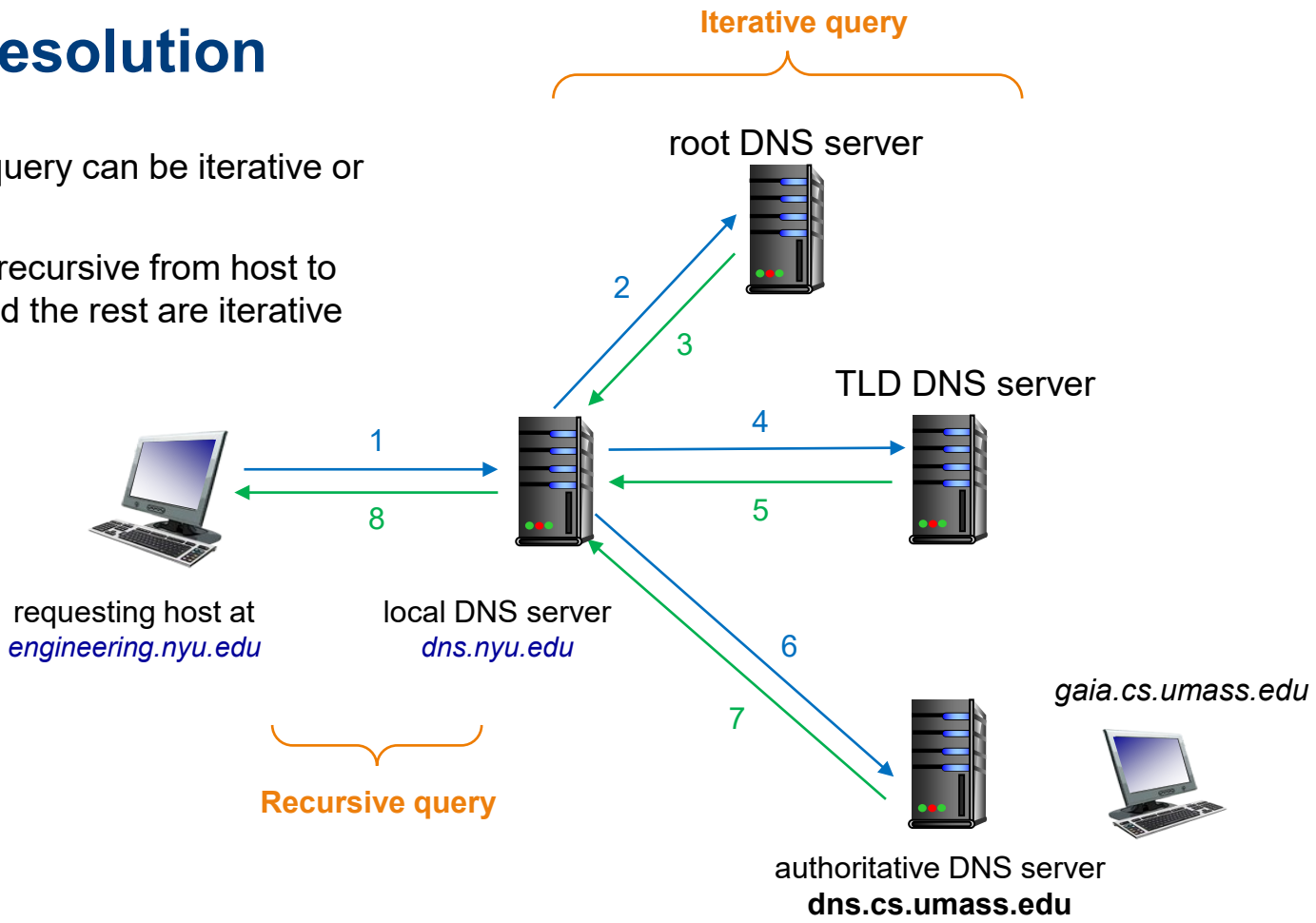
Recursive query:

- Puts burden of name resolution on contacted name server



DNS Name Resolution

- In theory, any DNS query can be iterative or recursive
- In practice, typically recursive from host to local DNS server, and the rest are iterative



Caching DNS Information

- Once (any) name server learns mapping, it *caches* the mapping, and immediately returns the cached mapping in response to a query
  - Improves response time
  - TLD servers are typically cached in local name servers
- Cache entries time-out after some time: time to live (TTL)

→ Cached entries may be *out-of-date*

  - If named host changes IP address, may not be known Internet-wide until all TTLs expire!
  - *Best-effort* name-to-address translation

- Aside: Your computer has its own caching too!
  - `ipconfig /displaydns`

DNS Records

DNS management for **examplesite1.cf**

+ Add record | Search DNS Records | Advanced

Type	Name	Content	TTL	Proxy status	
A	host	185.27.134.201	Auto	DNS only	Edit
A	examplesite1.cf	185.27.134.202	Auto	Proxied	Edit
A	www	185.27.134.202	Auto	Proxied	Edit
CNAME	mail	host.examplesite1.cf	Auto	DNS only	Edit
MX	examplesite1.cf	mail.examplesite1.cf	Auto	DNS only	Edit
TXT	examplesite1.cf	sample-txt-content	Auto	DNS only	Edit

DNS Records

DNS: distributed database storing *resource records* **(RR)**

RR format: (name, value, type, ttl)

type=A

- name is hostname
- value is IP address

type=NS

- name is domain (e.g., foo.com)
- value is hostname of authoritative name server for this domain

type=CNAME

- name is alias name for a canonical name
- value is canonical name

type=MX

- value is name of SMTP mail server associated with name

Others

- https://en.wikipedia.org/wiki/List_of_DNS_record_types

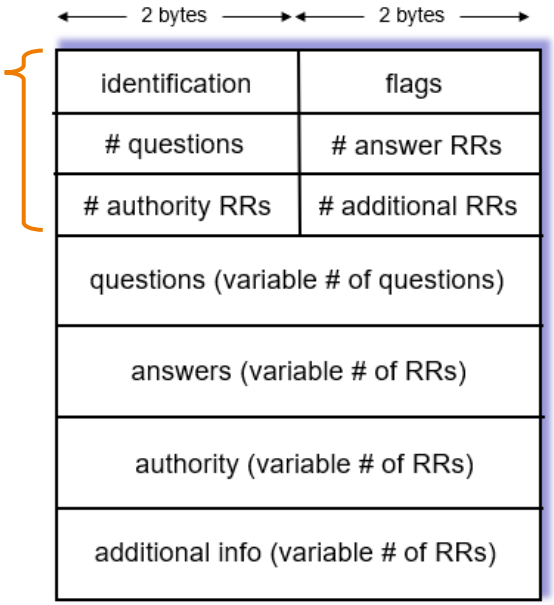
DNS Protocol Messages

- DNS *query* and *reply* messages have the same format:

Send a DNS query:
nslookup program

Message header

- **Identification**: 16-bit ID for query
  - Reply to query uses same ID
- **Flags:**
  - Query or reply
  - Reply is authoritative
  - Recursion desired
  - Recursion available



Name, type for a query

RRs in response to query

Records of other authoritative servers

Other helpful records

Domain Name Registrars

- Manage the reservation of Internet domain names
  - E.g., GoDaddy, NameCheap
- Typically also offer DNS hosting (i.e., runs DNS servers)
- Public users register their host name (e.g., networkuptopia.com) and IP address for resolution
- Registrar inserts appropriate NS, A RRs into .com TLD server

“Phonebook of the Internet”

- Humongous distributed database:
  - ~ billion records, each simple
- Handles many trillions of queries/day:
  - Many more reads than writes
  - Performance matters: almost every Internet transaction interacts with DNS - msecs count!
- Organizationally, physically decentralized:
  - Millions of different organizations responsible for their records
  - “Bulletproof”: reliability, security

Illustration of scale

- Comcast DNS servers alone: 600B DNS queries/day
- Akamai DNS servers alone: 2.2T DNS queries/day

Implementation

- DNS servers are often UNIX machines running the Berkeley Internet Name Domain (BIND) software
- Runs over UDP, using port 53

BBC NEWS

Major websites hit by global outage

22 July 2021

Many popular websites fell offline on Thursday in a widespread global outage of service.

Visitors attempting to reach some sites received DNS errors, meaning their requests could not reach the websites.

Affected services included Airbnb, UPS, HSBC bank, British Airways and the PlayStation network used for online games.

One popular DNS provider, Akamai, reported "an emerging issue" with its Edge DNS service.

It has now tweeted that the issue has been fixed and "the service is resuming normal operations".

Internet outage monitoring platform DownDetector had reported thousands of problems from its users across dozens of platforms.

The Internet is down.... It was DNS, again

adrian cockcroft · Follow · 4 min read · Oct 20, 2025

> *the main errors we are seeing are Could not resolve host: dynamodb.us-east-1.amazonaws.com*

In this most recent failure it was the DNS endpoint for DynamoDB in the AWS us-east-1 region that wasn't working. From the current status page:

> ***Oct 20 2:01 AM PDT*** *We have identified a potential root cause for error rates for the DynamoDB APIs in the US-EAST-1 Region. Based on our investigation, the issue appears to be related to DNS resolution of the DynamoDB API endpoint in US-EAST-1. We are working on multiple parallel paths to accelerate recovery. This issue also affects other AWS Services in the US-EAST-1 Region. Global services or features that rely on US-EAST-1 endpoints such as IAM updates and DynamoDB Global tables may also be experiencing issues. During this time, customers may be unable to create or update Support Cases.*

The problem with this particular failure is that lots of internal AWS services use DynamoDB to store their information, so many additional services broke, as well as several customers. Amazon retail, Amazon Alexa, Lyft, Snapchat, Signal etc. Even customers that are not hosted in us-east-1 would be affected indirectly as they could not create support cases, or change configurations of Identity and Access Management (IAM).